

MAHABAD, Iran, Islamic Republic of

WMO: 407260

Lat: 36.7529N

Lon: 45.7153E

Elev: 1352

StdP: 86.10

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.0	-8.8	-15.8	1.1	-8.5	-13.7	1.4	-5.1	15.2	5.0	13.8	5.0	0.0	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	36.8	16.2	35.2	16.0	33.8	16.0	18.6	31.8	17.9	31.0	17.3	30.2	5.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	11.5	24.8	12.8	10.8	24.7	12.0	10.3	24.3	58.4	31.7	56.1	31.0	54.0	30.1	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.2	8.9	DB	-14.5	39.7	3.1	1.1	-16.7	40.5	-18.5	41.1	-20.2	41.7	-22.5	42.4	(4)
(5)			WB	-15.8	19.6	2.7	1.0	-17.7	20.4	-19.3	21.0	-20.9	21.6	-22.8	22.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec				
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)				
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	0.2	3.1	8.2	12.6	18.1	23.4	26.6	26.1	21.7	15.2	7.7	2.3	(6)		
(7)		DBStd	9.77	4.72	4.16	4.15	3.42	2.69	2.31	2.28	2.12	2.64	3.70	3.97	4.88	(7)		
(8)		HDD10.0	929	303	194	83	14	0	0	0	0	0	5	89	240	(8)		
(9)		HDD18.3	2442	561	427	313	174	38	1	0	0	4	107	319	496	(9)		
(10)		CDD10.0	2333	0	1	29	92	252	402	516	501	352	168	20	2	(10)		
(11)		CDD18.3	804	0	0	1	2	32	153	257	242	107	12	0	0	(11)		
(12)		CDH23.3	9673	0	0	2	28	326	1712	3169	3062	1246	129	0	0	(12)		
(13)		CDH26.7	4607	0	0	0	3	57	723	1689	1639	479	16	0	0	(13)		
(14)	Wind	WSAvg	3.2	3.1	3.4	4.5	3.7	3.8	3.3	3.1	2.7	2.8	2.9	2.5	2.7	(14)		
(15)	Precipitation	PrecAvg	365	46	38	55	64	32	5	2	1	3	26	47	42	(15)		
(16)		PrecMax	612	79	119	101	132	102	39	13	11	21	121	135	111	(16)		
(17)		PrecMin	225	15	2	12	0	0	0	0	0	0	0	0	2	(17)		
(18)		PrecStd	94	19	25	23	34	26	8	4	2	5	33	36	27	(18)		
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.4	14.3	22.9	26.4	30.2	35.6	39.3	38.5	35.0	28.6	20.7	15.6	(19)		
(20)			MCWB	4.4	5.8	9.9	13.1	14.1	15.3	16.6	16.4	15.4	13.1	9.6	7.2	(20)		
(21)		2%	DB	9.4	12.3	20.1	23.6	28.2	33.7	37.3	36.7	32.7	26.6	17.9	12.8	(21)		
(22)			MCWB	3.9	5.6	8.9	11.9	13.9	15.2	16.4	16.1	14.8	12.4	8.8	5.7	(22)		
(23)		5%	DB	7.8	10.5	17.3	21.5	26.8	32.2	35.6	35.2	31.1	24.7	15.9	10.6	(23)		
(24)			MCWB	3.4	4.9	8.0	11.3	13.7	14.8	16.2	16.0	14.4	11.9	8.1	5.0	(24)		
(25)		10%	DB	6.5	9.1	15.2	19.6	25.1	30.7	33.9	33.9	29.5	22.7	14.1	8.7	(25)		
(26)			MCWB	2.6	4.3	7.2	10.5	13.2	14.7	16.2	16.2	14.0	11.5	7.8	4.2	(26)		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	7.7	10.6	13.9	17.1	18.0	19.4	19.5	17.4	14.8	11.5	8.2	(27)		
(28)			MCDWB	9.2	12.2	19.7	23.0	26.1	31.1	33.1	32.5	30.3	24.9	16.8	12.9	(28)		
(29)		2%	WB	4.7	6.3	9.5	12.9	15.4	17.0	18.7	18.6	16.2	13.7	10.4	6.6	(29)		
(30)			MCDWB	8.3	10.6	17.9	21.6	24.7	29.9	32.0	31.8	29.1	23.0	14.9	11.2	(30)		
(31)		5%	WB	3.7	5.3	8.6	12.1	14.6	16.3	18.1	17.9	15.5	12.9	9.4	5.5	(31)		
(32)			MCDWB	7.2	9.7	16.5	19.4	24.1	29.0	31.1	31.2	28.1	21.7	13.8	9.7	(32)		
(33)		10%	WB	2.8	4.4	7.6	11.2	13.8	15.7	17.5	17.3	14.8	12.2	8.3	4.5	(33)		
(34)			MCDWB	6.0	8.6	14.5	18.1	23.3	28.0	30.4	30.4	27.2	20.5	13.0	8.2	(34)		
(35)	Mean Daily Temperature Range			MDBR	7.3	8.4	9.6	10.9	12.2	13.7	13.5	14.8	14.9	12.1	9.5	7.2	(35)	
(36)		5% DB			MCDBR	8.5	10.6	12.5	14.2	14.3	15.8	16.0	17.0	17.1	14.9	12.2	9.4	(36)
(37)					MCWBR	4.9	5.6	5.5	6.0	5.5	5.5	5.4	5.9	6.6	6.0	6.1	5.2	(37)
(38)		5% WB			MCDWB	7.3	9.4	10.8	12.0	13.4	14.1	13.4	14.5	15.4	12.7	9.2	8.2	(38)
(39)					MCWBR	4.6	5.3	5.3	5.5	5.4	5.5	5.5	6.2	6.5	5.7	4.6	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.308	0.339	0.387	0.454	0.450	0.407	0.436	0.405	0.362	0.381	0.327	0.305	(40)		
(41)		taud		2.349	2.263	2.137	1.972	2.019	2.174	2.095	2.183	2.297	2.199	2.356	2.388	(41)		
(42)		Ebn at Noon		878	887	872	830	839	874	845	865	888	826	846	857	(42)		
(43)		Edh at Noon		93	115	143	179	173	148	159	143	120	121	92	84	(43)		
(44)	All-Sky Solar Radiation	RadAvg		2.50	3.43	4.36	5.46	6.82	8.26	7.94	7.28	6.03	4.12	2.79	2.17	(44)		
(45)		RadStd		0.24	0.35	0.41	0.47	0.41	0.34	0.19	0.23	0.32	0.31	0.37	0.29	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	-1.80	N/A	+1.04	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page